

RIDE BOOKING DATASET - DETAILED ANALYSIS REPORT

POWER BI

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SUMMARY:

This detailed report consolidates findings from a power BI analysis of a ride booking dataset containing 150,000 records and 21 columns. It covers data quality, cleaning, descriptive statistics, customer and driver behaviour, operational metrics (VTAT/CTAT), cancellation patterns, peak demand analysis, and recommendations for operational and analytical improvements.

KEY HIGHLIGHTS:

- Strong representation of 'auto' bookings (37,419) and lowest volume for 'uber XL' (4,449).**
- Average booking value is ₹508.30 and average ride distance is 24.64 km.**
- Cancellation is material: drivers cancelled ~27,000 rides (dataset-level number provided by source).**
- Cancellation peaks at ~10:00 AM and ~6:00 PM, with Monday being the highest-cancellation weekday.**
- No obvious correlation observed between booking value, distance and customer satisfaction in the provided analysis.**

1. Project Background & Objectives

Project: Analysis of Ride Booking Dataset (Power BI)

Primary objectives:

- Clean and standardize the dataset for reliable reporting.
- Produce descriptive analysis for vehicle types, distances, booking value and ratings.
- Surface customer cancellation behaviour and driver cancellations.
- Evaluate operational KPIs: VTAT (vehicle turnaround time) and CTAT (customer turnaround time).
- Identify peak demand locations and time slots to inform capacity planning.

2. Data & Methods

2.1 Data source and schema

The dataset contains 150,000 booking records and 21 columns including Date, Time, Booking ID, Booking Status, Customer ID, Vehicle Type, Pickup/Drop Location, VTAT, CTAT, cancellation flags + reasons, booking value, ride distance, ratings, and payment method.

2.2 Data cleaning & transformation (Power Query steps)

- Removed blank rows, removed rows with Power Query errors, and removed duplicate bookings.
- Replaced nulls in binary/flag columns (Cancelled Rides by Customer, Cancelled Rides by Driver, Incomplete Rides) with 0 to create clean binary flags.
- Converted numeric types where needed (e.g., changed decimal numbers to whole numbers for flags).
- Trimmed whitespace on text columns and standardized case for IDs (uppercased Booking ID & Customer ID) so 'ABC1234' and 'abc1234' are treated the same.
- Merged Date + Time into a single DateTime column and changed its type to Date/Time.
- Verified and corrected data types across all columns (dates, numbers, text).

2.3 DAX measures and derived columns (examples used):

- Day Name = FORMAT('Table'[DateTime], "dddd") -- weekday names (Monday..Sunday)
- HourOfDay = HOUR('Table'[DateTime]) -- integer 0..23
- Total Cancellation (measure) = SUM('Table'[Cancelled Rides by Customer]) + SUM('Table'[Cancelled Rides by Driver]) + SUM('Table'[Incomplete Rides])

2.4 Notes on methodology and assumptions

- The analysis uses the cleaned dataset exported from Power BI. All aggregates and charts referenced were produced inside the Power BI model.
- When the same location appears with slightly different text (e.g., spelling/casing), it may create duplicate location buckets; standardization is recommended.
- Driver-level performance metrics cannot be computed without a driver identifier in the source dataset — this is a key limitation.

3. Descriptive Findings

3.1 Vehicle type popularity

Most and least popular vehicle types (absolute counts provided by source analysis):

Vehicle Type	Bookings (count)	Notes
Auto	37,419	Most popular
Uber XL	4,449	Least popular
Premier Sedan	—	volume not provided in source summary

Visualization used: Bar chart (ranked by booking counts).

3.2 Average ride metrics

- Average booking value: ₹508.3
- Average ride distance: 24.64 km

Visualization: KPI cards in Power BI.

3.3 Rating distributions

Rating bins (counts provided):

Rating Bin	Count (Driver & Customer combined where noted)
4.0	43,901
4.5	25,713
5.0	2,365
3.5	14,327
3.0	6,694

Visualization used: Clustered column chart with rating bins.
Note: ensure bins match rating granularity (0.5 steps).

3.4 Cancellation reasons (top categories)

By	Top Reason	Count
Customer	Wrong address	2,362
Driver	Customer Related Issue	6,837

Visualization used: Two separate bar charts for customer vs driver cancellation reasons.

4. Customer Behaviour Insights

4.1 Frequent cancellers (by Customer ID)

- CID5481002 — 3 cancellations
- CID6715450 — 3 cancellations

Visualization used: Table showing Customer ID, total bookings, cancellations, cancellation rate.

4.2 Time-of-day & day-of-week cancellation patterns

- Peak cancellation times: ~10:00 AM and ~6:00 PM.
- Peak cancellation day: Monday.

Visualization used: Line chart (hourly cancellations) and bar chart (weekday totals).

4.3 Behavioural insights and operational recommendations (assistant additions)

- Implement a pre-ride confirmation SMS/push for bookings in the 9–11 AM and 5–7 PM windows to reduce no-shows and address-related cancellations.
- Flag customers with repeated cancellations and route them to a manual review or temporary hold with a short message explaining business rules.
- Add cancellation reason standardized categories and enforce structured reason reporting (dropdown) in driver and customer apps to reduce free-text variability.
- Introduce lightweight friction (e.g., small in-app deposit, penalty credit) only for repeat offenders — use A/B testing before wide rollout.

5. Driver Performance Evaluation

5.1 Per-driver ratings:

- Not available: driver identifiers were not present in the dataset, therefore individual driver ratings and acceptance rates cannot be computed. Collecting a unique Driver ID is recommended for future driver performance analysis.

5.2 Driver cancellations

- Total rides cancelled by drivers (from source): 27,000
- Most common driver cancellation reason: Customer Related Issue — 6,837

Operational recommendations (assistant additions):

- Track driver-level metrics: acceptance rate, cancellations per 100 rides, time-to-accept, and average on-trip time.
- Introduce targeted incentives in low-supply windows/locations to reduce 'No driver found' events (e.g., guaranteed hourly earnings for 5–7 PM shifts).
- Provide drivers with clearer structured cancellation reason options to improve root-cause analysis.

7. Correlation & Statistical Notes

The existing analysis reports 'no correlation' between ride value, ride distance and customer satisfaction. To validate and quantify this properly, the recommended approach is:

- Compute Pearson correlation for linear relationships and Spearman rank correlation to capture monotonic relationships.
- Visualize with scatter plots and add LOESS/smoothed trend lines to detect non-linear patterns.
- Control for confounders by segmenting (e.g., by vehicle type, location, and time of day) before correlational analysis.

Assistant addition:

When performing correlation tests, report both correlation coefficients and p-values, and consider effect sizes — statistically insignificant correlations can still be operationally meaningful in large samples.

8. Recommendations & Next Steps

Below are prioritized opportunities split into Quick Wins, Medium-term, and Strategic initiatives. (No time estimates provided; prioritize based on business needs.)

Quick Wins

- Standardize location naming using a reference table / fuzzy matching to collapse duplicate location entries.
- Enforce structured cancellation reasons (select lists) in driver/customer apps and backfill mappings for existing free-text reasons.
- Create KPI tiles in Power BI for: Daily cancellations, No-driver-found rate, Completed rides, and Active driver supply.
- Add a small 'Confirm pickup address' step in the app flow for bookings during peak cancellation hours.

Medium-term

- Instrument driver IDs in the dataset and create a driver performance dashboard (acceptance, cancellations, rating trends).
- Build a cancellation risk model (binary classification) using features such as customer history, time of day, pickup location, booking value, and vehicle type.
- Set up an alerting pipeline (Power BI data alerts or Power Automate) for sudden spikes in cancellations or 'no driver found' events.
- Segment customers by lifetime cancellations & booking frequency, and design retention tactics for high-value customers.

Strategic initiatives

- Develop a supply-demand forecasting model (time series + external features) to proactively incentivize drivers.
- Run experiments (A/B tests) for interventions: pre-ride confirmations, micro-deposits, or targeted driver incentives to measure impact on cancellations.
- Implement a real-time driver dispatch optimization layer to minimize 'no-driver-found' and reduce pickup delays.

KPIs to monitor (suggested):

- Cancellation rate (by driver/customer)
- No-driver-found rate
- Completed ride conversion rate
- Average VTAT/CTAT
- Driver acceptance rate
- Customer repeat booking rate
- Average booking value by segment

9. Limitations

- Driver identifiers are missing — limits ability to compute per-driver KPIs and attribute cancellations to drivers specifically.
- Potential duplicate/noisy location labels — affects location-level aggregates and peak-location identification.
- Free-text cancellation reasons increase classification noise; structured reasons reduce ambiguity.
- Reported 'no correlation' statement should be validated statistically with correlation coefficients and significance testing.

GitHub Repository Link :

https://github.com/pratikghag356/Ride_Booking_Analysis_Power-BI

Thank You